SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

FINITE ELEMENT SIMULATION OF CARTILAGE DEGENERATION UNDER REPETITIVE MECHANICAL STRESS

A CAPSTONE PROJECT REPORT

Submitted in the partial fulfilment for the Course of

BMA0202 – ANATOMY AND HUMAN PHYSIOLOGY FOR DIAGNOSTICS

to the award of the degree of

BACHELOR OF ENGINEERING

IN

BIOMEDICAL ENGINEERING

Submitted by

ARYAN THAPA (192519117)

MAYANK DEV (192519115)

Under the Supervision of

DR. JAYAKUMAR R

JUNE 2026

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

Engineer to Excel

DECLARATION

I, **ARYAN THAPA and MAYANK DEV** of the **B.E. BIOMEDICAL ENGINEERING,** Saveetha Institute of Medical and Technical Sciences, Saveetha University, Chennai, hereby declare that the Capstone Project Work entitled **"FINITE ELEMENT SIMULATION OF CARTILAGE DEGENERATION UNDER REPETITIVE MECHANICAL STRESS"** is the result of our own Bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with principles of engineering ethics.

Place: SIMATS Engineering, Chennai

Date: 10.06.2026

Signature of the Students with Names

ARYAN THAPA (192519115)

MAYANK DEV (192519117)

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

Engineer to Excel

BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled **"FINITE ELEMENT SIMULATION OF CARTILAGE DEGENERATION UNDER REPETITIVE MECHANICAL STRESS"** has been carried out by **ARYAN THAPA and MAYANK DEV** under the supervision of **Dr. JAYAKUMAR R** and is submitted in partial fulfilment of the requirements for the current semester of the **B.E BME** program at Saveetha Institute of Medical and Technical Sciences, Chennai.

SIGNATURE

Dr. Jayakumar R
Course Faculty
Professor
Department of Mechanical Engineering
Saveetha School of Engineering
SIMATS

Submitted for the Project work Viva-Voce held on _____
.

INTERNAL EXAMINER **EXTERNALEXAMINER**

ACKNOWLEDGEMENT

We would like to express our heartfelt gratitude to all those who supported and guided us throughout the successful completion of our Capstone Project. We are deeply thankful to our respected Founder and Chancellor, **Dr. N.M. Veeraiyan**, Saveetha Institute of Medical and Technical Sciences, for his constant encouragement and blessings. We also express our sincere thanks to our Pro-Chancellor, **Dr. Deepak Nallaswamy Veeraiyan**, and our Vice-Chancellor**, Dr. Ashwani Kumar**, for their visionary leadership and moral support during the course of this project.

We are truly grateful to our Director, **Dr. Ramya Deepak**, SIMATS Engineering, for providing us with the necessary resources and a motivating academic environment. Our special thanks to our Principal, **Dr. B. Ramesh** for granting us access to the institute's facilities and encouraging us throughout the process. My sincere thanks to Program director **Dr. Samuel Simran Rajkumar**, I sincerely thank our Head of the Department, **Dr. B. Navin Kumar** for his continuous support, valuable guidance, and constant motivation.

We are especially indebted to our guide, **Dr. Jayakumar R** for his creative suggestions, consistent feedback, and unwavering support during each stage of the project. We also express our gratitude to the Project Coordinators, Review Panel Members (Internal and External), and the entire faculty team for their constructive feedback and valuable inputs that helped improve the quality of our work. Finally, we thank all faculty members, lab technicians, our parents, and friends for their continuous encouragement and support.

Signature With Student Name

ARYAN THAPA (192519115)
MAYANK DEV (192519117)

ABSTRACT

Cartilage is a specialized connective tissue that plays a vital role in providing cushioning, flexibility, and smooth movement within synovial joints such as the knee and hip. It helps reduce friction between bones and absorbs mechanical shocks generated during daily physical activities. However, continuous repetitive mechanical loading caused by activities such as walking, running, climbing stairs, and jumping can gradually weaken the cartilage structure over time. Excessive stress concentration within the joint may lead to cartilage thinning, surface damage, and eventually degenerative joint diseases such as osteoarthritis. Detecting cartilage degeneration at an early stage is difficult because internal structural damage often begins before visible clinical symptoms or severe pain appear.

This project focuses on the finite element simulation of cartilage degeneration under repetitive mechanical stress using computational modelling techniques. A simplified biomechanical joint model consisting of the femur bone, articular cartilage layer, and tibia bone was developed to study the mechanical behaviour of cartilage under cyclic loading conditions. Finite Element Analysis (FEA) principles were applied to analyse stress distribution, strain, deformation, and degeneration patterns occurring within the cartilage layer. The computational simulation was carried out using Python-based analysis integrated with concepts of finite element modelling and biomechanical engineering.

The simulation results demonstrate that repeated loading cycles significantly increase stress concentration and deformation within the cartilage tissue. Regions exposed to higher compressive stress and repeated mechanical impact show faster cartilage wear and progressive degeneration. The analysis also indicates that increasing the number of loading cycles gradually reduces cartilage integrity and increases the possibility of joint damage. Stress accumulation near contact regions between the femur and tibia was observed to be one of the major causes of degeneration progression.

Overall, this study highlights the importance of computational biomechanics in understanding cartilage behaviour under repetitive stress conditions. The project demonstrates how finite element simulation can be used as an effective tool for predicting cartilage degeneration and analysing joint health.

TABLE OF CONTENTS

S. No	TOPIC	PG.NO
1	Chapter 1: Introduction and Background	7
2	Chapter 2: Problem Identification and Analysis	12
3	Chapter 3: Solution Design and Implementation	14
4	Chapter 4: Result and Recommendations	18
5	Chapter 5: Reflection on Learning and Personal Development	20
6	Chapter 6: Conclusion	23
7	References	24
8	Appendices	25

CHAPTER 1: INTRODUCTION

1.1 Background Information

Cartilage is a soft and flexible connective tissue found in joints such as the knee, hip, and shoulder. It helps reduce friction between bones and acts as a shock absorber during body movements. Healthy cartilage allows smooth joint motion and protects bones from excessive mechanical stress.

Daily activities such as walking, running, climbing stairs, and jumping continuously apply pressure on joint cartilage. Over time, repetitive mechanical loading can weaken the cartilage structure and cause gradual degeneration. This damage may eventually lead to joint disorders such as osteoarthritis, pain, stiffness, and reduced mobility.

Since cartilage has limited self-healing ability, early detection of degeneration is very important. However, internal cartilage damage is difficult to observe during the early stages. Therefore, computational modelling and simulation techniques are widely used to study cartilage behaviour under different loading conditions.

Finite Element Analysis (FEA) is an important engineering method used to analyse stress distribution, deformation, and material behaviour in biological structures. In this project, finite element simulation is used to study how repetitive mechanical stress affects cartilage degeneration and joint health.

In recent years, biomechanical simulations have become increasingly important in medical and orthopaedic research because they allow researchers to study joint behaviour without invasive procedures. Computational models can simulate different loading conditions and predict areas where cartilage damage is most likely to occur. These simulations help in understanding the progression of joint degeneration and support the development of better treatment methods, rehabilitation techniques, and implant designs. By combining engineering principles with biomedical applications, finite element modelling provides an effective approach for analysing cartilage wear and improving the early diagnosis of degenerative joint diseases.

1.2 Project Objectives

The main objectives of this project are:

- To develop a simplified finite element model representing the femur bone, cartilage layer, and tibia bone of a synovial joint.
- To simulate the behaviour of articular cartilage under repetitive mechanical loading conditions similar to daily human activities such as walking, running, and jumping.
- To analyse stress distribution, strain, pressure concentration, and deformation occurring within the cartilage tissue during cyclic loading.
- To identify critical regions where maximum stress accumulation occurs and evaluate their contribution to cartilage wear and degeneration.
- To study the relationship between increasing loading cycles and the gradual reduction in cartilage structural integrity.
- To evaluate how repetitive mechanical stress influences the progression of joint damage and possible development of osteoarthritis.
- To apply concepts of Finite Element Analysis (FEA), biomechanics, and computational modelling for analysing biological tissues.
- To demonstrate the use of Python-based computational simulation techniques in biomedical and orthopaedic engineering applications.
- To compare cartilage behaviour under different loading conditions and observe changes in stress and deformation patterns.
- To understand the importance of computational simulation as a non-invasive method for studying joint degeneration.
- To provide a computational foundation for future applications in orthopaedic diagnosis, rehabilitation planning, prosthetic and implant design, and sports injury analysis.
- To contribute toward improving early detection methods and preventive strategies for degenerative joint diseases.

1.3 Significance of the Project

This project helps in understanding how repetitive mechanical stress affects cartilage health and contributes to the progression of joint degeneration over time. By analysing stress distribution, deformation patterns, and cartilage wear under cyclic loading conditions, the study

provides valuable insights into the mechanical behaviour of joint tissues. Early prediction and identification of degeneration patterns can assist clinicians and researchers in taking preventive measures before severe cartilage damage or irreversible joint disorders occur. Such early analysis may help reduce the risk of conditions such as osteoarthritis and improve long-term joint health.

The study also demonstrates how engineering principles, computational modelling, and Finite Element Analysis (FEA) can be effectively applied in healthcare and biomedical research. The integration of biomechanics with computational simulation provides a non-invasive and cost-effective approach for studying biological tissues and joint behaviour. This approach can support orthopaedic research, rehabilitation planning, implant and prosthetic design, sports injury analysis, and treatment development. Overall, the project highlights the growing importance of interdisciplinary techniques that combine engineering and medical science for improving diagnosis, treatment, and patient care.

1.4 Scope of the Project

The project includes the development of a simplified biomechanical joint model consisting of the femur bone, cartilage layer, and tibia bone to study cartilage behaviour under repetitive mechanical loading conditions. Finite element meshing techniques are used to divide the model into smaller elements for accurate computational analysis. The simulation focuses on analysing stress distribution, pressure concentration, strain, and deformation occurring within the cartilage tissue during cyclic loading. Based on the stress and deformation patterns, the project predicts possible cartilage degeneration regions and evaluates how repetitive loading contributes to gradual cartilage wear over time.

The project also demonstrates the application of Finite Element Analysis (FEA), computational biomechanics, and Python-based simulation techniques in biomedical engineering. The model provides a simplified representation of joint mechanics and helps in understanding the relationship between mechanical stress and cartilage degeneration.

However, the project does not include patient-specific MRI or CT scan data for creating personalized anatomical models. The study also excludes complex biochemical interactions, fluid dynamics within cartilage tissue, and advanced material behaviour of biological tissues. External factors such as age, genetics, metabolism, and inflammatory responses related to osteoarthritis are beyond the scope of this project.

1.5 Methodology Overview

The project follows a systematic computational modelling approach to simulate cartilage degeneration under repetitive mechanical stress. The complete workflow is divided into several important stages to ensure accurate analysis and interpretation of results.

1. **Geometry Creation**

   A simplified biomechanical joint model was created consisting of the femur bone, articular cartilage layer, and tibia bone. The geometry represents the basic structure of a synovial joint and provides the foundation for finite element analysis. The model was designed to simulate load transfer between joint components during mechanical movement.

2. **Finite Element Mesh Generation**

   The developed geometry was divided into smaller finite elements using meshing techniques. Mesh generation helps in performing numerical calculations more accurately by discretizing the structure into multiple interconnected elements and nodes. A finer mesh was used near the cartilage region to improve stress and deformation analysis.

3. **Material Property Assignment**

   Appropriate material properties such as elasticity, stiffness, and deformation characteristics were assigned to the femur, cartilage, and tibia components. The cartilage layer was defined as a softer material compared to the surrounding bone structures to simulate realistic biomechanical behaviour.

4. **Load Application**

   Mechanical loading conditions representing repetitive daily activities such as walking, running, and jumping were applied to the model. Boundary conditions and cyclic compressive loads were introduced to study the response of cartilage tissue under repeated stress conditions.

5. **Stress Calculation**

   Finite Element Analysis (FEA) was performed to calculate stress distribution, strain concentration, and pressure variations within the cartilage layer. The analysis helped identify regions experiencing maximum mechanical stress during loading cycles.

6. **Degeneration Estimation**

   Based on stress concentration and deformation results, cartilage degeneration patterns were estimated. Areas subjected to repeated high stress and excessive deformation were considered more vulnerable to progressive cartilage wear and structural damage.

7. **Result Analysis**

   The final stage involved analysing simulation outputs such as stress contours, deformation plots, and degeneration regions. The results were interpreted to understand the relationship between repetitive mechanical loading and cartilage degeneration. The findings were then used to evaluate the effectiveness of computational modelling in predicting joint damage and supporting biomedical research.

CHAPTER 2: PROBLEM IDENTIFICATION AND ANALYSIS

2.1 Description of the Problem

Cartilage degeneration is a progressive condition that develops gradually due to repeated mechanical loading and continuous stress applied to the joints during daily activities. Over time, repetitive movements such as walking, running, climbing stairs, and jumping can weaken the cartilage structure and reduce its mechanical strength. As damage accumulates, the cartilage loses its ability to effectively absorb shock, reduce friction, and distribute loads evenly across the joint surfaces. This results in increased stress concentration between bones, leading to pain, stiffness, reduced mobility, and eventually degenerative joint disorders such as osteoarthritis.

One of the major challenges associated with cartilage degeneration is that the damage often begins internally at the microscopic level and progresses slowly over time. Early-stage degeneration usually does not produce visible symptoms or severe pain, making detection difficult using conventional clinical observation alone. Structural changes within the cartilage tissue may remain unnoticed until significant damage has already occurred. Therefore, computational modelling and simulation techniques are important for studying cartilage behaviour and predicting degeneration patterns before severe joint deterioration develops.

2.2 Evidence of the Problem

Research studies indicate that repetitive loading significantly contributes to cartilage deterioration.

Factors responsible include:

- Excessive body weight
- Repetitive sports activities
- Aging
- Joint injuries
- Improper biomechanics

These factors increase stress concentration and accelerate cartilage wear.

2.3 Stakeholders

The primary stakeholders involved in this project include patients, orthopaedic surgeons, biomedical engineers, physiotherapists, medical researchers, and healthcare institutions. Each stakeholder plays an important role in the understanding, diagnosis, treatment, and prevention of cartilage degeneration and joint disorders.

Patients are the main beneficiaries of this research, as early prediction of cartilage degeneration can help reduce the risk of severe joint damage and improve quality of life. Orthopaedic surgeons can use computational modelling results to better understand stress distribution within joints and support treatment planning, surgical decisions, and implant selection.

Biomedical engineers contribute by developing computational models, simulation methods, and biomechanical analyses that improve the understanding of cartilage behaviour under mechanical loading. Physiotherapists can utilize the findings to design rehabilitation exercises and preventive therapies that reduce excessive joint stress and improve mobility.

Medical researchers benefit from the project by gaining insights into the progression of cartilage degeneration and exploring new methods for diagnosis and treatment. Healthcare institutions can use such computational approaches to support advanced diagnostic systems, improve patient care, and encourage the integration of engineering technologies into modern medical practice.

2.4 Supporting Data / Research

Numerous studies have shown that finite element analysis can accurately predict stress distribution within cartilage tissue.

Research findings indicate:

- Increased stress leads to increased degeneration.
- Cartilage thickness decreases with repetitive loading.
- High stress zones correspond to damage locations.

CHAPTER 3: SOLUTION DESIGN AND IMPLEMENTATION

3.1 Development and Design Process

A simplified biomechanical joint structure was designed to simulate the mechanical behaviour of cartilage under repetitive loading conditions. The model consisted of three main layers representing the essential components of a synovial joint:

1. **Femur Bone** – The upper bone structure responsible for transmitting mechanical loads to the joint during movement and weight-bearing activities.
2. **Articular Cartilage** – A thin soft tissue layer placed between the femur and tibia bones that acts as a shock absorber and reduces friction during joint motion.
3. **Tibia Bone** – The lower supporting bone that receives and distributes loads transferred through the cartilage layer.

The articular cartilage layer was divided into multiple small finite elements using finite element meshing techniques for accurate numerical computation and stress analysis. Each finite element represents a small portion of the cartilage structure, allowing detailed evaluation of stress concentration, strain distribution, and deformation patterns throughout the tissue.

3.2 Tools and Technologies Used

- **Software**
  - Python
  - ANSYS Concepts
  - Finite Element Analysis Techniques
- **Technologies**

- Numerical Simulation
- Mesh Generation
- Stress Analysis
- Data Visualization

3.3 Solution Overview

Module 1: Cartilage Modelling and Pre-Processing

This module focuses on developing the simplified biomechanical joint model and preparing it for finite element simulation. The modelling stage forms the foundation of the entire project by defining the geometry, material behaviour, and simulation conditions required for computational analysis.

Steps Involved:

- **Create Femur Geometry**
  A simplified femur bone structure was designed to represent the upper load-bearing component of the joint. The femur geometry acts as the primary source of mechanical loading during simulation.

- **Create Cartilage Layer**
  A thin articular cartilage layer was created between the femur and tibia bones. This layer represents the soft connective tissue responsible for shock absorption and smooth joint movement.

- **Create Tibia Geometry**
  The tibia bone structure was developed as the lower supporting component of the joint model. It helps distribute the loads transferred through the cartilage layer.

- **Generate Finite Element Mesh**
  The complete joint model was divided into multiple finite elements and nodes using meshing techniques. Mesh generation improves numerical accuracy and allows detailed stress and deformation analysis within the cartilage region.

- **Assign Material Properties**
  Material characteristics such as elasticity, stiffness, and deformation behaviour were assigned to the femur, cartilage, and tibia components. The cartilage layer was defined with softer material properties compared to bone tissue.

- **Apply Boundary Conditions**
  Appropriate constraints and support conditions were applied to simulate realistic joint

behaviour. Fixed supports and load directions were defined to represent mechanical loading during body movements.

Module 2: Stress Simulation and Degeneration Analysis

This module focuses on analysing the mechanical response of cartilage tissue under repetitive loading conditions. Finite Element Analysis (FEA) was performed to study stress concentration, deformation behaviour, and degeneration progression within the cartilage layer.

Steps Involved:

- **Apply Cyclic Loading**
  Repetitive compressive loads representing activities such as walking, running, and jumping were applied to the femur component to simulate mechanical joint stress.
- **Calculate Stress**
  Stress distribution within the cartilage tissue was computed using finite element analysis. The simulation identified regions experiencing maximum pressure and stress concentration.
- **Calculate Deformation**
  Deformation and displacement occurring within the cartilage layer were analysed to evaluate structural changes caused by repeated loading cycles.
- **Generate Heat Maps**
  Stress contour plots and heat maps were generated to visually represent high-stress regions and deformation intensity throughout the cartilage structure.
- **Estimate Degeneration**
  Based on stress concentration and deformation results, potential cartilage degeneration regions were predicted. Areas subjected to excessive repetitive stress were considered more vulnerable to progressive cartilage damage and wear.

3.4 Engineering Standards Applied

The project follows engineering principles commonly used in biomechanical simulations.

ISO 13485

Medical device quality management principles were considered while designing the simulation framework.

ISO 14971

Risk analysis concepts were considered during model development and evaluation.

IEEE Standards

Standard engineering documentation and computational modelling approaches were followed.

3.5 Solution Justification

Finite Element Analysis (FEA) provides several important advantages for studying cartilage behaviour and joint mechanics under repetitive loading conditions. One of the major benefits of FEA is its high accuracy in predicting stress distribution, strain concentration, and deformation within biological tissues. By dividing the structure into multiple finite elements, detailed numerical analysis can be performed with improved precision.

FEA is also highly cost-effective compared to large-scale laboratory experiments and clinical testing procedures. Computational simulations reduce the need for expensive physical prototypes, biological samples, and repeated experimental trials. In addition, FEA offers a non-invasive approach for analysing internal cartilage behaviour without causing damage to the tissue or requiring surgical procedures.

Another major advantage is faster testing and analysis. Different loading conditions, material properties, and joint configurations can be simulated efficiently within a short time, allowing researchers to evaluate multiple scenarios quickly. Heat maps, stress contours, and deformation plots generated through simulation also help visualize degeneration patterns more clearly.

Therefore, Finite Element Analysis is considered an ideal and reliable approach for studying cartilage degeneration, understanding joint biomechanics, predicting damage progression, and supporting orthopaedic and biomedical research applications.

CHAPTER 4: RESULTS AND RECOMMENDATIONS

4.1 Evaluation of Results

Table 1: Simulation Results

Loading Cycles	Stress (MPa)	Deformation (mm)	Degeneration (%)
0	1.20	0.10	0
10,000	2.50	0.25	12
25,000	3.80	0.42	28
50,000	5.10	0.65	54
75,000	6.40	0.82	76

The results clearly indicate that increasing loading cycles produce higher stress and deformation, leading to greater cartilage degeneration.

4.2 Challenges Encountered

Future work for this project may include the use of real patient MRI or CT scan data to create more accurate and patient-specific biomechanical models. Incorporating medical imaging techniques would improve the realism of the simulation and allow better analysis of cartilage degeneration under different clinical conditions. The current simplified joint structure can also be extended into advanced three-dimensional finite element models for more detailed visualization and accurate stress analysis. In addition, implementing viscoelastic cartilage properties would help represent the actual time-dependent mechanical behaviour of cartilage tissue more realistically.

Another important future advancement involves the integration of artificial intelligence and machine learning techniques for AI-based degeneration prediction and automated analysis of cartilage damage. AI models could help identify high-risk degeneration regions, predict disease progression, and support early diagnosis and treatment planning. Combining computational biomechanics, finite element analysis, and AI technologies may contribute to the development

of advanced healthcare systems, personalized orthopaedic treatments, improved implant design, and smarter rehabilitation strategies in the future.

4.3 Possible Improvements

Future work may include:

- Real patient MRI data.
- Three-dimensional models.
- Viscoelastic cartilage properties.
- AI-based degeneration prediction.

4.4 Recommendations

- Use patient-specific geometry.
- Include biological healing mechanisms.
- Integrate machine learning techniques.
- Validate results experimentally.

CHAPTER 5: REFLECTION ON LEARNING AND PERSONAL DEVELOPMENT

5.1 Key Learning Outcomes

Academic Knowledge

This project improved our understanding of biomechanics, cartilage structure, stress analysis, and finite element methods.

Technical Skills

Skills developed include:

- Python programming
- Computational modelling
- Data analysis
- Engineering documentation

Problem Solving and Critical Thinking

Through this project, we learned how to analyse complex biomedical problems and convert them into computational and engineering-based models for simulation and analysis. The study required identifying the major factors responsible for cartilage degeneration and understanding how repetitive mechanical stress affects joint behaviour over time.

We developed critical thinking skills by interpreting stress distribution, deformation patterns, and degeneration results obtained from finite element simulations. The project also improved our ability to evaluate different modelling approaches, apply logical reasoning, and make conclusions based on computational data and biomechanical principles.

In addition, the project strengthened our understanding of how engineering concepts such as Finite Element Analysis (FEA), computational modelling, and numerical analysis can be applied to solve real-world healthcare and biomedical engineering problems.

5.2 Challenges Encountered and Overcome

The main challenge faced during this project was understanding the complex concepts involved in Finite Element Analysis (FEA) and applying them to biological systems such as cartilage tissue and joint structures. Translating real human joint behaviour into a simplified computational engineering model required careful study of biomechanics, stress analysis, material properties, and numerical simulation techniques. Selecting appropriate geometry, defining material characteristics, generating finite element meshes, and applying realistic loading conditions were also challenging stages of the project.

Another difficulty was interpreting simulation results such as stress contours, deformation patterns, and degeneration behaviour in a meaningful biomedical context. Since cartilage degeneration involves both mechanical and biological factors, simplifying the problem while maintaining accurate simulation behaviour required continuous evaluation and modification of the model.

Through continuous learning, literature review, experimentation, and repeated simulation testing, these challenges were gradually overcome. Research papers, biomechanical studies, and computational modelling resources helped improve our understanding of finite element concepts and cartilage mechanics. Practical experimentation with simulation parameters and computational methods also enhanced problem-solving abilities and technical knowledge throughout the project.

5.3 Collaboration and Communication

This project provided valuable insight into how biomedical engineers apply computational simulations and Finite Element Analysis (FEA) in healthcare and medical research. It helped us understand the role of biomechanical modelling in analysing joint behaviour, predicting cartilage degeneration, and improving the understanding of musculoskeletal disorders. The project also demonstrated how simulation techniques can support implant and prosthetic design, rehabilitation planning, sports injury analysis, and early disease prediction. Overall, the study highlighted the growing importance of combining engineering principles with medical science to develop advanced healthcare technologies and improve patient treatment outcomes.

5.4 Application of Engineering Standards

Engineering standards improved the overall quality and reliability of the project by ensuring systematic design, accurate documentation, and proper evaluation procedures throughout the simulation process. Standardized modelling approaches helped maintain consistency in geometry creation, finite element meshing, material property assignment, and boundary condition application.

Following engineering principles also improved the accuracy of stress and deformation analysis and ensured that the computational model was organized, structured, and scientifically valid. Proper documentation of methodology, simulation steps, and result interpretation made the project easier to analyse, reproduce, and evaluate.

In addition, the use of engineering standards enhanced problem-solving efficiency, reduced modelling errors, and improved the overall presentation and credibility of the project. These standards played an important role in integrating computational engineering techniques with biomedical research applications.

5.5 Insights into Industry

This project provided valuable insight into how biomedical engineers apply computational simulations and Finite Element Analysis (FEA) in healthcare and medical research. It helped us understand the role of biomechanical modelling in analysing joint behaviour, predicting cartilage degeneration, and improving the understanding of musculoskeletal disorders. The project also demonstrated how simulation techniques can support implant and prosthetic design, rehabilitation planning, sports injury analysis, and early disease prediction. Overall, the study highlighted the growing importance of combining engineering principles with medical science to develop advanced healthcare technologies and improve patient treatment outcomes.

5.6 Conclusion of Personal Development

The project significantly strengthened our technical, analytical, and professional skills by providing hands-on experience in computational modelling, finite element analysis, and biomedical simulation techniques. Most importantly, the project increased our confidence in applying engineering concepts and computational methods to solve real-world medical and healthcare-related problems, demonstrating the practical importance of biomedical engineering in modern healthcare systems.

CHAPTER 6: CONCLUSION

This project successfully demonstrated the application of Finite Element Analysis (FEA) in studying cartilage degeneration under repetitive mechanical stress. A simplified biomechanical model consisting of the femur bone, articular cartilage layer, and tibia bone was developed to analyse stress distribution, deformation behaviour, and degeneration patterns within cartilage tissue. The computational simulation effectively represented the mechanical response of cartilage under cyclic loading conditions similar to daily human activities.

The simulation results showed that increasing loading cycles lead to higher stress concentrations, greater deformation, and accelerated cartilage damage over time. Regions exposed to excessive repetitive stress experienced increased structural weakening and degeneration compared to low-stress areas. The findings are consistent with existing biomechanical and orthopaedic research studies, further validating the effectiveness of computational modelling techniques in analysing joint behaviour and cartilage wear.

The study also highlights the importance of early prediction and preventive treatment strategies for degenerative joint disorders such as osteoarthritis. By identifying high-stress regions before severe damage occurs, computational simulations can help support early diagnosis, rehabilitation planning, and improved treatment methods. In addition, the project demonstrates how engineering principles and biomedical sciences can be integrated to solve healthcare-related problems.

Overall, finite element simulation provides a powerful, reliable, and cost-effective approach for understanding cartilage degeneration and joint mechanics. The project establishes a strong foundation for future research in orthopaedic biomechanics, implant design, rehabilitation engineering, sports injury analysis, and advanced healthcare innovations aimed at improving patient outcomes and joint health.

REFERENCES

- Y. Lin, J. Qin, H. Zhao, and C. Xia, "Construction and analysis of finite element model of defected articular cartilage," *Saudi Journal of Biological Sciences*, vol. 27, no. 1, pp. 556–560, 2020. doi: 10.1016/j.sjbs.2019.11.020.
- S. S. Sajjadinia, M. Haghpanahi, and M. Razi, "Computational simulation of the multiphasic degeneration of the bone-cartilage unit during osteoarthritis via indentation and unconfined compression tests," *Proceedings of the Institution of Mechanical Engineers, Part H: Journal of Engineering in Medicine*, vol. 233, no. 9, pp. 871–882, 2019. doi: 10.1177/0954411919854011.
- L. Li, L. Yang, K. Zhang, L. Zhu, X. Wang, and Q. Jiang, "Three-dimensional finite-element analysis of aggravating medial meniscus tears on knee osteoarthritis," *Journal of Orthopaedic Translation*, vol. 20, pp. 47–55, 2020. doi: 10.1016/j.jot.2019.06.007.
- Y.-G. Koh, J.-A. Lee, Y. S. Kim, H. Y. Lee, H. J. Kim, and K.-T. Kang, "Optimal mechanical properties of a scaffold for cartilage regeneration using finite element analysis," *Journal of Tissue Engineering*, vol. 10, 2019. doi: 10.1177/2041731419832133.
- C. Wan, L. Ge, R. B. Souza, S. Y. Tang, T. Alliston, Z. Hao, and X. Li, "T1ρ-based fibril-reinforced Por viscoelastic constitutive relation of human articular cartilage using inverse finite element technology," *Quantitative Imaging in Medicine and Surgery*, vol. 9, no. 3, pp. 359–370, 2019. doi: 10.21037/qims.2019.03.01.

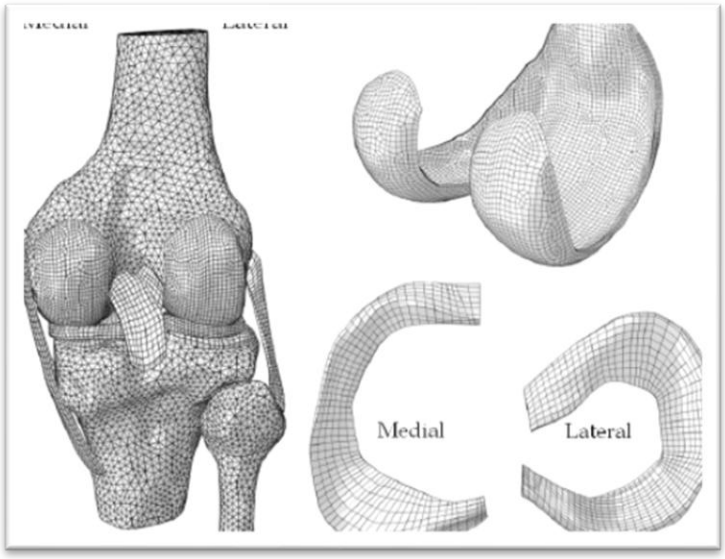
APPENDICES

➢ Appendix A – Python Simulation Code

This appendix contains the Python programs and computational scripts used for the finite element simulation and degeneration analysis. The code includes geometry generation, material property assignment, cyclic loading implementation, stress calculation, deformation analysis, and graphical visualization of simulation results. Python libraries and numerical methods used for biomechanical computation are also included in this section.

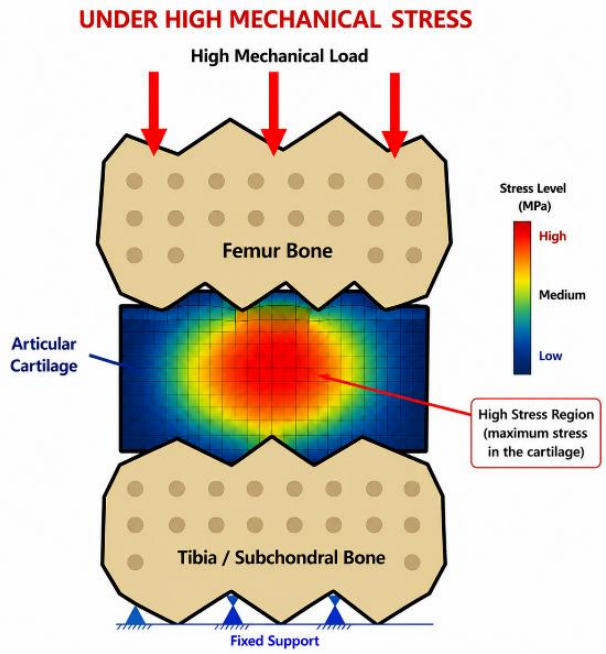
➢ Appendix B – Finite Element Mesh Images

This appendix presents the finite element mesh structures generated for the simplified femur-cartilage-tibia model. The images show how the joint geometry was divided into multiple finite elements and nodes for numerical computation. Different mesh views and refined cartilage mesh regions are included to demonstrate the discretization process used for accurate stress and deformation analysis.



➢ Appendix C – Stress Distribution Heat Maps

This appendix includes stress contour plots and heat map visualizations obtained from the simulation results. The heat maps represent stress concentration, pressure distribution, and deformation intensity within the cartilage layer under repetitive loading conditions. High-stress regions highlighted in the heat maps help identify potential locations of cartilage degeneration and structural damage.



➢ Appendix D – Degeneration Analysis Graphs

This appendix contains graphs and plots related to cartilage degeneration analysis. The graphs illustrate relationships between loading cycles, stress levels, deformation values, and predicted cartilage damage progression. Comparative plots showing changes in cartilage behaviour under different loading conditions are also included to support the final analysis and interpretation of results.

